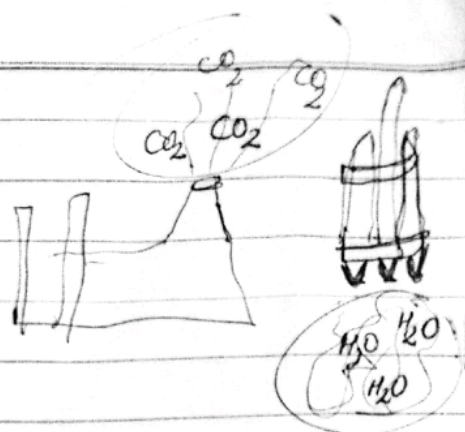


* Some components at inlet ^{that will be} different at the exit.
 $C_3H_8 + O_2$

Fuel + Oxygen $\xrightarrow{\text{Heat \& light}}$ Energy + other pro
Combustion reactions

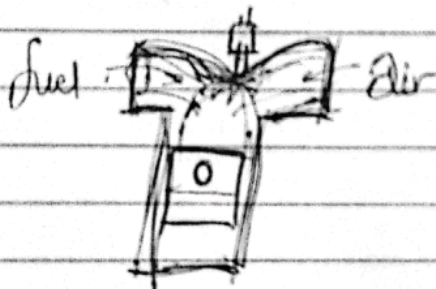
Coal + Oxygen $\rightarrow$ Carbon dioxide + energy

Hydrogen + Oxygen $\rightarrow$ Water + Energy

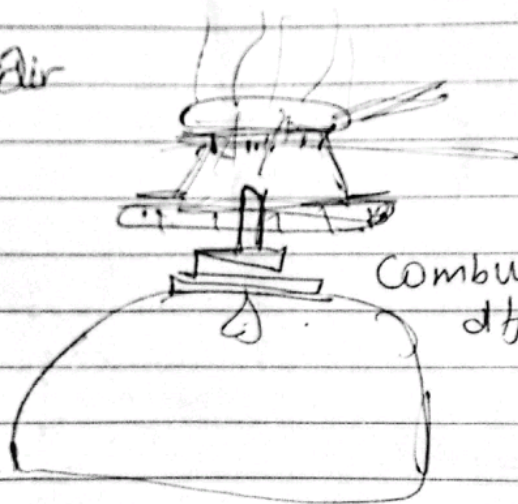


Some combustion rxns are very fast

Match

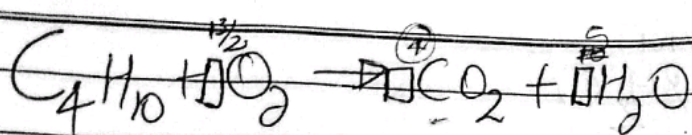


* Needs to be controlled.



Combustion at
diff rate

Chemical
products
will be diffg
to atmosphere



$$C = 4$$

$$H = 10$$

$$O = 2$$

$$C = 1 \times 4 = 4$$

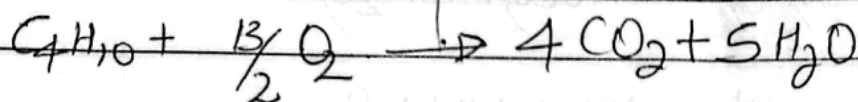
$$H = 2 \times 5 = 10$$

$$O = 2 + 1 = 3 \rightarrow 2 \times 4 + 5 = 13$$

To balance oxygen

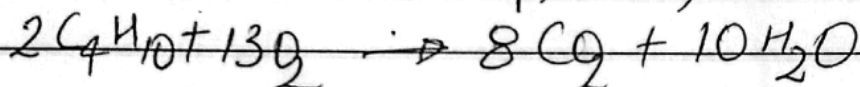
$$\begin{array}{r} 13 \\ 2 \times 13 \\ \hline 26 \\ \hline 13 \end{array}$$

$$O \rightarrow 13$$



to get whole no.

↳ multiply it by 2

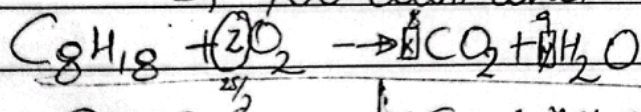


→ Remember to count all of the Oxygen atoms!

→ Balance C & H first. [Do O last]

→ Use a fraction to balance the O atoms. $\frac{13}{2}$

→ If you want whole numbers in the equation, multiply all of the coefficients by 2.



$$C = 8$$

$$H = 18$$

$$O = 2$$

$$C = 1 \times x = 8 \rightarrow x = 8$$

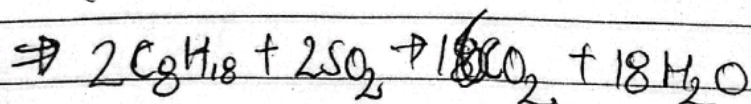
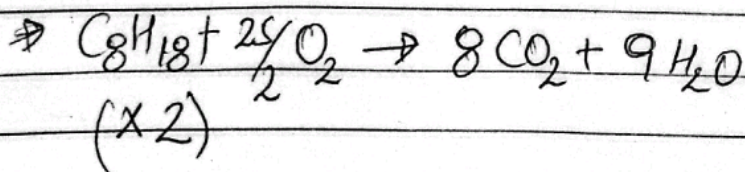
$$H = 2 \times y = 18 \rightarrow y = \frac{18}{2} = 9$$

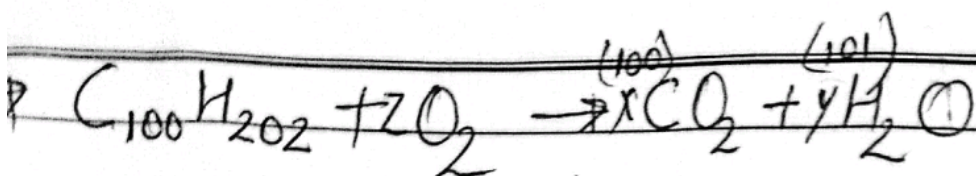
$$O = 2 + 1 = 3 \rightarrow 2 \times 8 + 9 = 25$$

$$O = 2 \times 2 = 4$$

$$Z = 25$$

$$2$$





$$C = 100$$

$$H = 202$$

$$O = 2$$

$$C = 1 \times x = 100 \Rightarrow x = 100$$

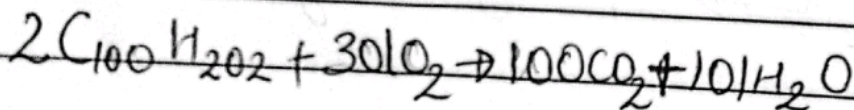
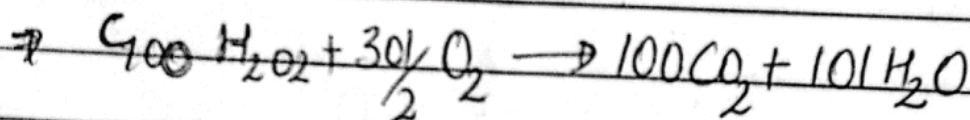
$$H = 2 \times y = 202 \Rightarrow y = \frac{202}{2} = 101$$

$$O = (2 + 1 \times 3) \Rightarrow 2 \times 100 + 101$$

$$= 301$$

$$\downarrow O = 2z = 301$$

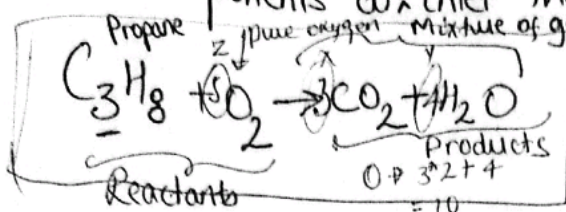
$$z = \left(\frac{301}{2} \right)$$



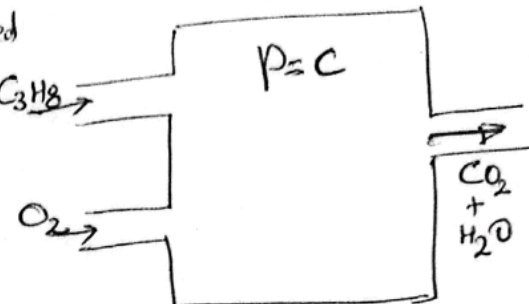
Combustion / Chemical Reaction

#3.1

→ Some components at inlet that will be different at the exit. Complete combustion



- Atoms are conserved
- No. of moles is not conserved



→ In earlier chapters: same components was in and out!

Now, reactants & products are different so, there is price to pay (energy) to destroy the reactants & price to pay (energy) to combine C & O₂ & H₂ to O to create water.

Destroying some structures and creating others
Energy is paid

Incomplete combustion
caused when there is no enough oxygen
if there is poor mixing (no very good mixing)

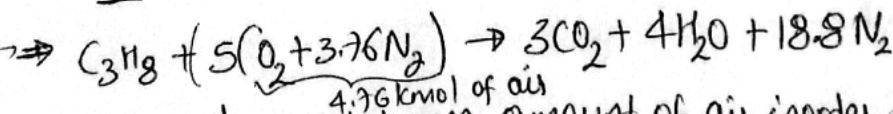
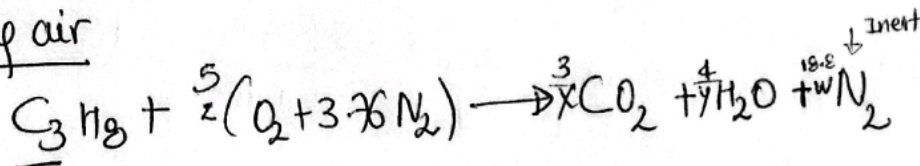
Pure Oxygen is not used in real life but air.

Air

$$\begin{matrix} 79\% \text{ N}_2 & \Rightarrow & \frac{79 \text{ kmol N}_2}{21 \text{ kmol O}_2} = 3.76 \frac{\text{kmol N}_2}{\text{kmol O}_2} \\ 21\% \text{ O}_2 & & \end{matrix}$$

For 1 kmol O₂ + 3.76 kmol N₂
4.76 kmol of air

Using air

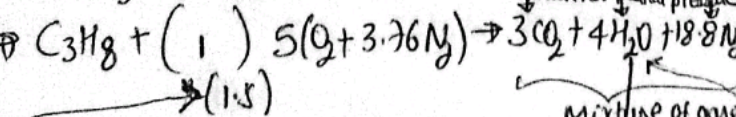


→ We need a minimum amount of air in order to have a complete combustion.
→ Theoretical air (Stoichiometric air)

→ Sometimes more air is needed/added

→ Excess air

E.g. - 50% excess air - 50% extra - 150% Theoretical air



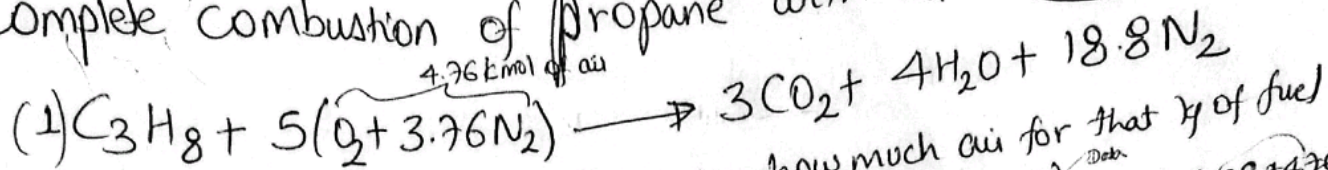
Why: 1. to ensure complete combustion
2. to control the temperature in the combustion chamber

Mixture of gases in vapor form: $T_{\text{dp}} = ?$

At $P=C$, when T drops, the start of condensation water

like $\text{SO}_2 \rightarrow \text{H}_2\text{SO}_4$ (sulfuric acid) which has corrosive nature
Partial pressure $\Rightarrow T_{\text{dp}} \Rightarrow P_{\text{dp}}$ which starts condensation
if T drops below $T_{\text{dp}} \Rightarrow$ water vapor starts to condensate at constant P

Complete combustion of Propane with air (#1.1)



Important characteristic: Air-Fuel ratio: how much air for that of fuel

$$AF = \frac{\text{Mass of Air}}{\text{Mass of fuel}} = \frac{(n \cdot M)_{\text{Air}}}{(n \cdot M)_{\text{fuel}}} = \frac{5 \cdot 29 \cdot 4.76}{1 \cdot (3 \cdot 12 + 8 \cdot 1)} = 15.69 \frac{\text{kg air}}{\text{kg } C_3H_8}$$

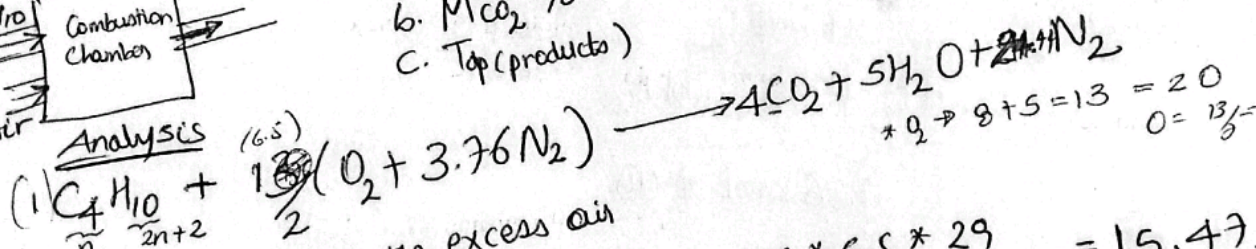
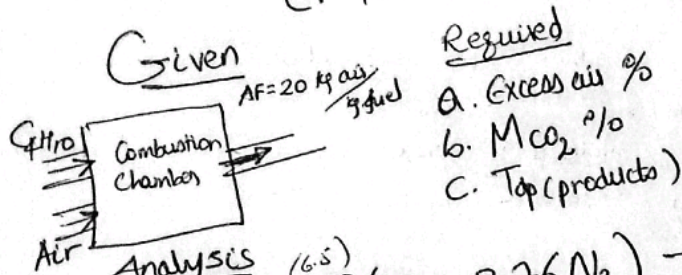
(*) If $AF > 15.69 \Rightarrow$ Lean mixture $\Rightarrow$ more air
 $AF < 15.69 \Rightarrow$ Rich mixture $\Rightarrow$ Less air (more fuel)

Example

Butane is burned with dry air-fuel ratio of 20. Calculate

- The percentage of excess air.
- The molar percentage of CO_2 in the products.
- The dew-point temp. of the products.

Assumption
 - Combustion is complete.
 - Combustion gases are ideal gases.



a. Check the excess air

$$AF = \frac{M_{\text{air}}}{M_{\text{fuel}}} = \frac{(n \cdot M)_{\text{air}}}{(n \cdot M)_{\text{fuel}}} = \frac{4.76 \cdot 6.5 \cdot 29}{1 \cdot (4 \cdot 12 + 10 \cdot 1)} = 15.47 \frac{\text{kg air}}{\text{kg fuel}}$$

$\Rightarrow 15.47 \frac{\text{kg air}}{\text{kg fuel}} \rightarrow 100\% \text{ theoretical air.}$

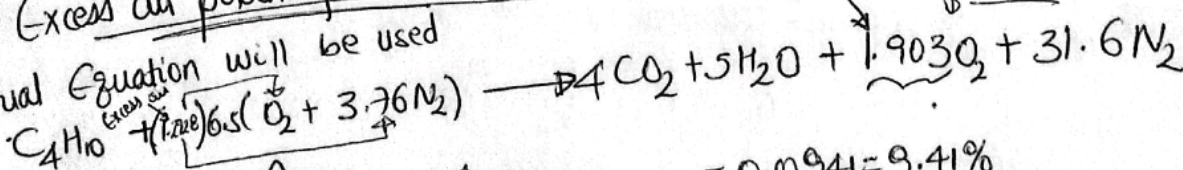
$$\Rightarrow ?\% = \frac{20 \cdot 100\%}{15.47} = 129.28\%$$

Given is $20 \frac{\text{kg air}}{\text{kg fuel}} \rightarrow ?\%$

$$= 29.28\% \text{ excess air}$$

$\Rightarrow$ Excess air percentage = 29.28%

b. Actual Equation will be used



Balance oxygen (O)

$\Rightarrow \% CO_2 = \frac{n_{CO_2}}{n_{\text{Total product}}} = \frac{4}{4 + 5 + 1.903 + 31.6} = 0.0941 = 9.41\%$

c. Top (products)

C_{dp} Tdp

⇒ It's referring to H_2O water vapor.
T at which condensation of water can happen at $p=c$
 $P_v = ?$

Dalton model = $\frac{P_i}{P_m} = y_i = n_i \Rightarrow$

$T_{dp}(\text{products}) = 48.86^\circ C$

⇒ If T_{products} drops below $48.86^\circ C$
we will end up with liquid water.
↳ Must be followed since water will react with other products.

$P_c = P_v = y_{H_2O} P_m$

Assume $P_m = 1 \text{ atm} = 101.325 \text{ kPa}$ (not given)

$y_{H_2O} = \frac{n_{H_2O}}{n_m} = \frac{5}{4+5+1.903+31.6} = 0.1176$

⇒ $P_v = 0.1176 \times 101.325 \text{ kPa} = 11.87 \text{ kPa}$

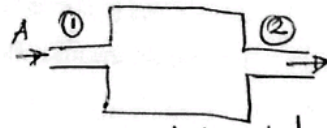
Go to table A-15

P	$T_{\text{sat}}^\circ C$
10	45.81
11.87	?
15	53.97

⇒ $\frac{? - 45.81}{53.97 - 45.81} = \frac{11.87 - 10}{15 - 10}$
⇒ $? = 48.86^\circ C$

Enthalpy of formation, combustion: In combustion, we are destroying or creating new components & there is energy associated with it.

Reference enthalpy for each component



Reference enthalpy
→ From table

$\Delta h = h_{2/A} - h_{1/A}$

$(h_{2/A} - h_{\text{ref,A}}) - (h_{1/A} - h_{\text{ref,A}})$

but it will be diff for combustion since product is different
 $\Delta h = h_{2/B} - h_{1/A}$

$= (h_{2/B} - h_{\text{ref,B}}) - (h_{1/A} - h_{\text{ref,A}})$
 ↓ ↓
 $h_{\text{ref,B}}$ $h_{\text{ref,A}}$

$\bar{h}$ is used for combustion b/c we care more abt Energy/kmol than Energy/kg

$\bar{h}^\circ |_{p=1 \text{ atm}, T=25^\circ C}$

Example $\Delta \bar{h}_{N_2} = \bar{h}_{N_2@500K} - \bar{h}_{N_2@25+273}^\circ$
 $= 14,581 \text{ kJ/kmol} - 9,669 \text{ kJ/kmol}$

Enthalpy of formation

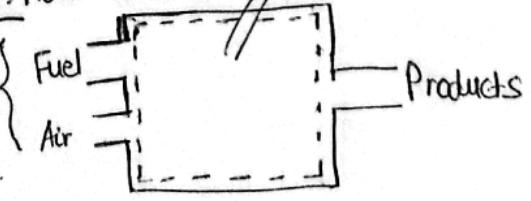
$\bar{h}_f^\circ$ - from table → energy released or used during destroying or creating in combustion

h_c - Heat released during a steady state combustion process. of fuel, a complete combustion heat is released at certain T & p.

** Most important question: how much heat we are getting? $Q?$

→ Applying 1st law of Thd to combustion:

Enthalpy = $\bar{h}_f^\circ + (\bar{h} - \bar{h}^\circ)$
(since both reactants & products are different)
 ↑ ↑
 Enthalpy of formation Actual Enthalpy



1st law, steady process: $\Delta PE + \Delta KE$ are negligible
 $\dot{Q}_{in} + \dot{W}_{in} + \sum n_r (\bar{h}_f^\circ + (\bar{h} - \bar{h}^\circ)) = \dot{Q}_{out} + \dot{W}_{out} + \sum n_p (\bar{h}_f^\circ + (\bar{h} - \bar{h}^\circ))$

Open system:

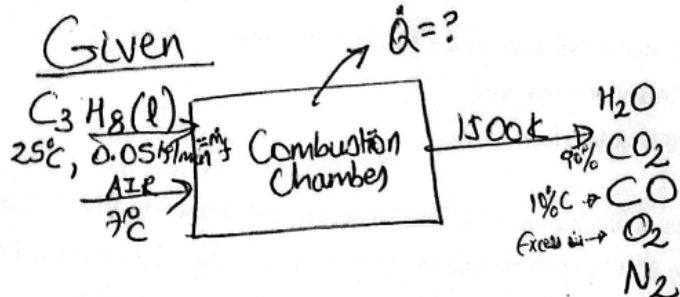
Closed system: 1st law eqn using internal energy (u) → $h = u + Pv \rightarrow u = h - Pv$
⇒ For closed system
 $\sum n_r (\bar{h}_f^\circ + (\bar{h} - \bar{h}^\circ - P_r v_r)) + \sum n_p (\bar{h}_f^\circ + (\bar{h} - \bar{h}^\circ - P_p v_p)) = 0$

Example Boles page 767

(#33)

Liquid propane (C_3H_8) enters a combustion chamber at $25^\circ C$ at a rate of 0.05 kg/min where it is mixed and burned with 50% excess air that enters the combustion chamber at $7^\circ C$, as shown in Fig. An analysis of the combustion gases reveals that all the hydrogen in the fuel burns to H_2O but only 90% of the carbon burns to CO_2 , with the remaining 10 percent forming CO . If the exit temperature of the combustion gases is $1500 K$, determine (a) the mass flow rate of air and (b) the rate of heat transfer from the combustion chamber.

Solution



Required

- $\dot{m}_{air}$
- $\dot{Q}_{out}$

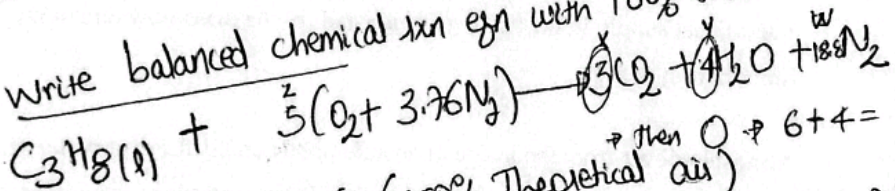
Assumptions

- Steady operation conditions exist.
- Air and the combustion gases are ideal gases.
- Kinetic & potential energies are negligible.

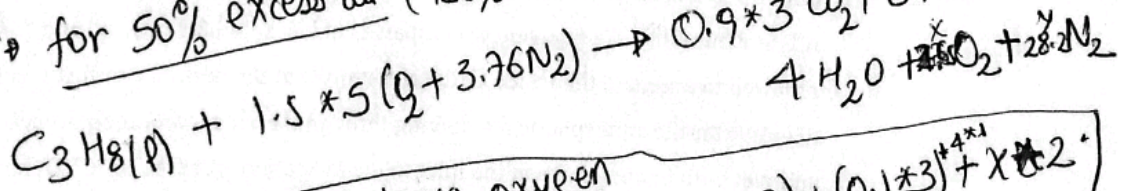
Analysis

a. $\dot{m}_{air}$

1st → Write balanced chemical rxn eqn with 100% theoretical air.



2nd → for 50% excess air (150% Theoretical air)



To balance N_2

$$1.5 \times 5 \times 3.76 = y N_2$$

$$y = 28.2$$

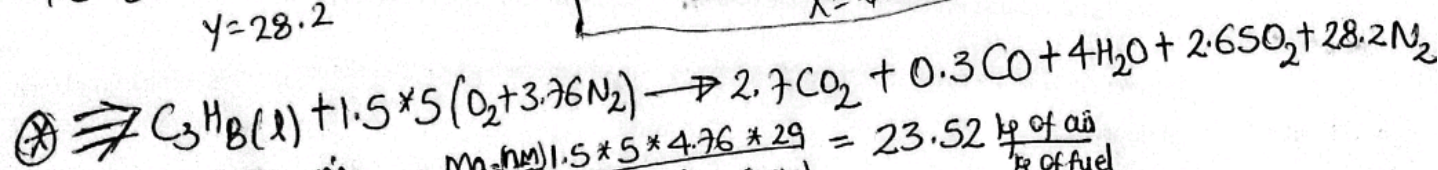
To balance oxygen

$$1.5 \times 5 \times 2 = (0.9 \times 3 \times 2) + (0.1 \times 3 \times 1) + x \times 2$$

$$15 = 5.7 + 2x + 4$$

$$15 = 9.7 + 2x$$

$$x = 2.65$$



SO, $AF = \frac{\dot{m}_{air}}{\dot{m}_{fuel}} = \frac{m_{air}(NM)_{in}}{m_{fuel}(NM)_{in}} = \frac{1.5 \times 5 \times 4.76 \times 29}{(3 \times 12) + (8 \times 1)} = 23.52 \frac{\text{kg of air}}{\text{kg of fuel}}$

Then $\dot{m}_{air} = (AF) \dot{m}_{fuel} = 23.52 \frac{\text{kg of air}}{\text{kg of fuel}} \times 0.05 \frac{\text{kg of fuel}}{\text{min}} = 1.176 \frac{\text{kg of air}}{\text{min}}$

↳ means for the chemical equation for 0.05 kg of fuel per minute, 1.176 kg of air is needed per minute.

b. $\dot{Q}_{out}$

Applying 1st law of THD
 $\dot{Q}_{in} + \dot{W}_{in} + \sum n_r (\bar{h}_{f,i} + \bar{h}_i - \bar{h}_i^0) = \dot{Q}_{out} + \dot{W}_{out} + \sum n_p (\bar{h}_{f,e} + \bar{h}_e - \bar{h}_i^0)$
 Entropy of formation
 Actual enthalpy
 Reference enthalpy

→ Best way is to use table:

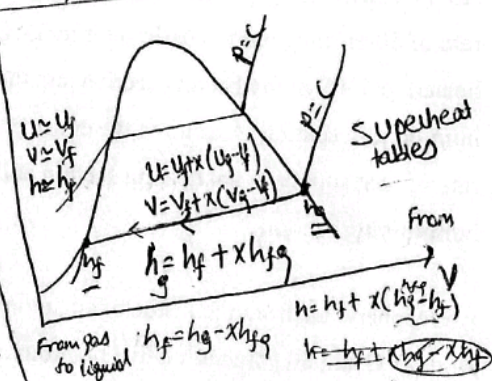
Temperatures
 Reactant: at 25°C - Fuel = 298K
 Product: at 1500K
 25°C = 298K
 1500K = 1500K

Substance	$\bar{h}_f^0$ (kJ/kmol)	$\bar{h}_{298K}$	$\bar{h}_{298K}$	$\bar{h}_{1500K}$	$\bar{h}_{298K}$
$C_3H_8(l)$	-103,850 - 335*44	-	-	-	-
O_2	0	8150	8682	49292	-
N_2	0	8141	8669	47023	-
CO_2	-393,520	-	9364	71078	-
CO	-110,530	-	8669	47517	-
H_2O	-241,820	-	9904	57999	-

→ For reactant:
 the state is at
 reference state 25°C
 $(\bar{h}_i - \bar{h}_i^0) = 0$

→ C_3H_8 (propane) is
 in liquid phase
 but $\bar{h}_f$ values on table (Table A-26)
 are in gaseous phase.

→ Oxygen is in the reactant
 & product.



Octane in liquid phase
 $h_f = h_{hg}$
 Table A-27
 $C_3H_8(l) \rightarrow C_3H_8(g)$
 $h_f^0 - h_{fg}$
 Entropy of vaporization

$$\dot{Q}_{out} = (1) (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 7.5 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 3.76 \times 7.5 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) - [2.7 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 0.3 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 4 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 2.65 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0) + 28.2 (\bar{h}_f^0 + \bar{h}_i - \bar{h}_i^0)]$$

but h_{fg} in Table (A-27) is in kJ/kg
 what we need is per kmol
 $m = n \times M$
 $n = \frac{m}{M}$
 $\bar{h}_f^0 |_{C_3H_8(g)} - M h_{fg}$
 $-103,850 \text{ kJ/kmol} - 44.097 \text{ kg/kmol} \times 335 \text{ kJ/kg}$

$$\Rightarrow \dot{Q}_{out} = \sum (\bar{h}_{f,i} + \bar{h}_i - \bar{h}_i^0) - \sum n_p (\bar{h}_{f,e} + \bar{h}_e - \bar{h}_i^0)$$

$$\dot{Q}_{out} = (1) (-103,850 - 335 \times 44) + 7.5 (8150 - 8682) + 3.76 \times 7.5 (8141 - 8669) - [2.7 (-393,500 + 71078 - 9364) + 0.3 (-110,530 + 47517 - 8669) + 4 (-241,820 + 57999 - 9904) + 2.65 (0 + 49292 - 8682) + 28.2 (0 + 47023 - 8669)] = 308,137.9 \text{ kJ/kmol fuel}$$

→ Adiabatic flame temperature

Assume you have a chemical reaction

↳ ~~What~~ is the maximum temp you can get from this chemical reaction?

Adiabatic mean

↳ All the energy is used to increase the enthalpy, nothing is lost.

Since its assumed as ideal gases

$$h = f(T)$$

$$\Rightarrow h \uparrow \Rightarrow T \uparrow$$

$$\Rightarrow \text{Adiabatic} \Rightarrow \dot{Q}_{out} + \sum_{P=i} n_{P,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) = \sum_{R=i} n_{R,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0)$$

$$\sum_{P=i} n_{P,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) = \sum_{R=i} n_{R,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0)$$

Annotations:
 - n : no of moles
 - $\bar{h}_f^0$: enthalpy of formation
 - $\bar{h}$: Actual enthalpy
 - $\bar{h}^0$: Reference enthalpy

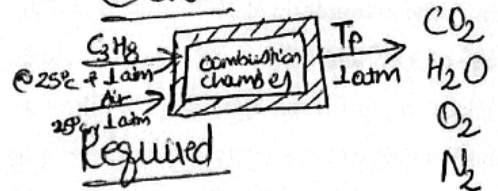
Find temp. which satisfies this eqn.

Example

Propane is burned with 250% theoretical air, both are at 25°C & 1 atm.

Predict the adiabatic flame temperature in the steady flow combustion chamber

Given

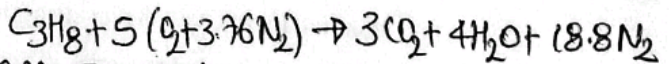
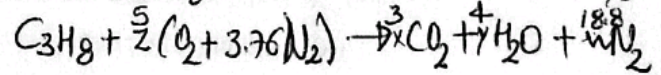


Required

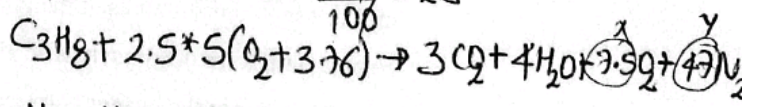
Adiabatic flame temp.?

Analysis

→ Chemical rxn eqn with theoretical air 100%



→ 250% Theoretical air → $\frac{250}{100} = 2.5$



Assumptions

1. This is steady flow combustion.
2. The combustion chamber is adiabatic.
3. There are no work interactions.
4. Air & the combustion gases are ideal gases.
5. Changes in kinetic & potential energies are negligible.

→ For Reactants

$$H_R = H_P \quad \text{1st Law of THD}$$

$$\left[n_{C_3H_8} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) + n_{O_2} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) + n_{N_2} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) \right]_{25^\circ C} = \left[n_{CO_2} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) + n_{H_2O} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) + n_{O_2} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) + n_{N_2} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) \right]_{T_P}$$

Annotations:
 - $T_{ref} = T_{act} = 298 K$
 - $\bar{h}_{actual} = \bar{h}_{ref} \Rightarrow \bar{h}_f$
 - $\bar{h}_{f,O_2} = \bar{h}_{f,N_2} = 0$
 - $n_{C_3H_8} = 1$

$$\bar{h}_f|_{C_3H_8} = 3(\bar{h}_f|_{CO_2} + \bar{h}_{CO_2,T_P} - \bar{h}_{CO_2}^0) + 4(\bar{h}_f|_{H_2O} + \bar{h}_{H_2O,T_P} - \bar{h}_{H_2O}^0) + 7.5(\bar{h}_f|_{O_2} + \bar{h}_{O_2,T_P} - \bar{h}_{O_2}^0) + 47(\bar{h}_f|_{N_2} + \bar{h}_{N_2,T_P} - \bar{h}_{N_2}^0)$$

→ Trick to reduce unknowns:

- When we see the eqn no of mole of N_2 is the largest/highest: N_2 dictates the temp.
 ↳ Assume the product only has nitrogen as unknown

→ Adiabatic flame temperature

Assume you have a chemical reaction

↳ What is the maximum temp you can get from this chemical reaction?

Adiabatic mean

↳ All the energy is used to increase the enthalpy, nothing is lost.

Since its assumed ideal gases

$$h = f(T)$$

$$\Rightarrow h \uparrow \Rightarrow T \uparrow$$

$$\Rightarrow \sum_{P=i} n_{P,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0) = \sum_{R=i} n_{R,i} (\bar{h}_{f,i}^0 + \bar{h}_i - \bar{h}_i^0)$$

Annotations: n = no of moles, h_f⁰ = enthalpy of formation, h = actual enthalpy, h⁰ = reference enthalpy

↳ Extremely important parameter since it helps 1. to design combustion chamber
↳ material to resist this temp.
2. Maximum efficiency

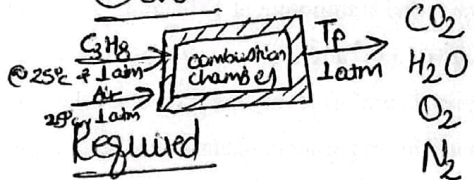
Find temp. which satisfies this eqn.

Example

Propane is burned with 250% theoretical air, both are at 25°C & 1 atm. Predict the adiabatic flame temperature in the steady flow combustion chamber

Solution

Given



Required

Adiabatic flame temp.?

Analysis

→ Chemical rxn eqn with theoretical air 100%

$$C_3H_8 + 5(O_2 + 3.76 N_2) \rightarrow 3CO_2 + 4H_2O + 18.8 N_2$$

$$C_3H_8 + 5(O_2 + 3.76 N_2) \rightarrow 3CO_2 + 4H_2O + 18.8 N_2$$

 → 250% Theoretical air → $\frac{250}{100} = 2.5$

$$C_3H_8 + 2.5 \times 5(O_2 + 3.76) \rightarrow 3CO_2 + 4H_2O + 7.5O_2 + 47N_2$$

Assumptions

1. This is steady flow combustion.
2. The combustion chamber is adiabatic.
3. There are no work interactions.
4. Air & the combustion gases are ideal gases.
5. Changes in kinetic & potential energies are negligible.

⇒ For Reactants

$H_R = H_P$ 1st Law of THD

$$\left[n_{C_3H_8} (\bar{h}_{f,C_3H_8}^0 + \bar{h}_{C_3H_8} - \bar{h}_{C_3H_8}^0) + n_{O_2} (\bar{h}_{f,O_2}^0 + \bar{h}_{O_2} - \bar{h}_{O_2}^0) + n_{N_2} (\bar{h}_{f,N_2}^0 + \bar{h}_{N_2} - \bar{h}_{N_2}^0) \right]_{T_{ref}} = \left[n_{CO_2} (\bar{h}_{f,CO_2}^0 + \bar{h}_{CO_2} - \bar{h}_{CO_2}^0) + n_{H_2O} (\bar{h}_{f,H_2O}^0 + \bar{h}_{H_2O} - \bar{h}_{H_2O}^0) + n_{O_2} (\bar{h}_{f,O_2}^0 + \bar{h}_{O_2} - \bar{h}_{O_2}^0) + n_{N_2} (\bar{h}_{f,N_2}^0 + \bar{h}_{N_2} - \bar{h}_{N_2}^0) \right]_{T_p}$$

Annotations: T_{ref} = T_{ad} = 298 K, h_{actual} = h_{ref} ⇒ h_f, h_{f,O₂} = h_{f,N₂} = 0, n_{C₃H₈} = 1

$$\bar{h}_f|_{C_3H_8} = 3(\bar{h}_f|_{CO_2} + \bar{h}_{CO_2} - \bar{h}_{CO_2}^0) + 4(\bar{h}_f|_{H_2O} + \bar{h}_{H_2O} - \bar{h}_{H_2O}^0) + 7.5(\bar{h}_f|_{O_2} + \bar{h}_{O_2} - \bar{h}_{O_2}^0) + 47(\bar{h}_f|_{N_2} + \bar{h}_{N_2} - \bar{h}_{N_2}^0)$$

⇒ Trick to reduce unknowns:

- when we see the eqn no of mole of N₂ is the largest/highest: ⇒ N₂ dictates the temp.
 ↳ Assume the product only has nitrogen as unknown

$$\# \bar{h}_f^0|_{C_3H_8} = 3(\bar{h}_f^0|_{CO_2} + \bar{h}_{CO_2@T_p} - \bar{h}_{CO_2}^0) + 4(\bar{h}_f^0|_{H_2O} + \bar{h}_{H_2O@T_p} - \bar{h}_{H_2O}^0) + 7.5(\bar{h}_f^0|_{O_2} + \bar{h}_{O_2@T_p} - \bar{h}_{O_2}^0) + 47(\bar{h}_f^0|_{N_2} + \bar{h}_{N_2@T_p} - \bar{h}_{N_2}^0) \quad (\#36)$$

Only 1 unknown $\bar{h}_{N_2}$

From table

	$\bar{h}_f^0$	$\bar{h}_{298K}^0$
C_3H_8	-103850	-
CO_2	-393520	9360
$H_2O(g)$	-241820	9900
O_2	0	8680
N_2	0	8670

$$\Rightarrow -103850 = 3(-393520 + \bar{h}_{N_2} - 9360) + 4(-241820 + \bar{h}_{N_2} - 9900) + 7.5(0 + \bar{h}_{N_2} - 8680) + 47(\bar{h}_{N_2} - 8670)$$

$$= (3\bar{h}_{N_2} + 4\bar{h}_{N_2} + 7.5\bar{h}_{N_2} + 47\bar{h}_{N_2}) -$$

$$\bar{h}_{N_2} = 42,020.4878 \rightarrow \text{table} \rightarrow T = 1380K$$

→ This is not the correct since only N_2 is considered.

SO: $H_p|_{@1380K} \rightarrow$ Now $\bar{h}_{CO_2}@1380K$ from table & use in the Eqn

$\bar{h}_{H_2O}@1380K$

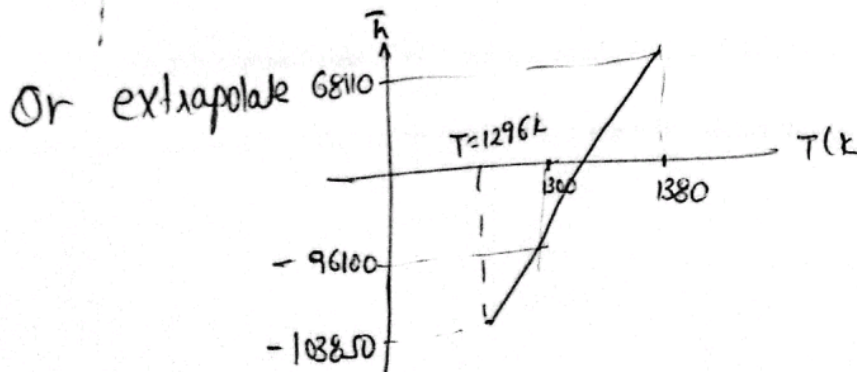
$\bar{h}_{O_2}@1380K$

$\bar{h}_{N_2}@1380K$

$$H_p|_{@1380K} = 68110 \text{ kJ/kmol} \neq H_R = (-103850 \text{ kJ/kmol})$$

↳ SO not the correct temp.

Iterate: take $T = 1300K \Rightarrow H_p = -96100 \text{ kJ/kmol} \neq H_R$



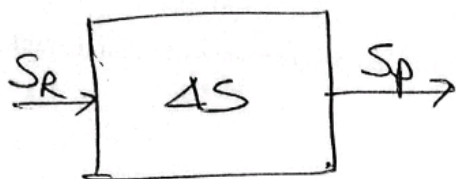
⇒ Adiabatic flame temp is 1296K ⇒ $T = 1296K$

Entropy Generation ^{& Most beautiful property}

(#3.7)

↳ 1. Exergy destruction

2. Computation of reversible work / the maximum work we can get



Balance S

$$S_{in} - S_{out} + S_{gen} = \Delta S / \text{system}$$

$$\sum \frac{Q}{T} + S_{gen} = S_{product} - S_{reactant}$$

Assume the system as adiabatic: $\sum \frac{Q}{T} \rightarrow 0$

$$S_{gen} = S_{product} - S_{reactant}$$

Find common reference for entropy

↳ 3rd law of T.H.D: "The entropy of pure crystalline substance at zero absolute Temp is 'zero'"

$$S|_{T=0K} = 0$$

$$S|_{T=0K} = 0$$

⊗ Entropy unlike h & u is dependent on 2 properties T & p

$$S = S(T, p)$$

But product is a mixture of ideal gases: Dalton's law.

Partial Pressure:

↳ Each component's entropy will be evaluated by its own partial pressure.

⇒ To make calculations easier

↳ Entropy at reference value (@ 1 atm) & correct it by adding term.

$$S_i(T_i, P_i) = \underbrace{\bar{S}_i^0(T, P=1\text{atm})}_{\text{Table}} - R_u \ln \left(\underbrace{\frac{y_i P_m}{P_0 @ 1\text{atm}}}_{\text{Partial pressure of each component}} \right)$$

Example (Cengel Boles: page 777)

Methane (CH_4) gas enters a steady-flow adiabatic combustion chamber @ 25°C & 1 atm. It is ~~and~~ burned with 50% excess air that also enters at 25°C & 1 atm. Assuming complete combustion and an adiabatic flame temp. of 1789K, determine the entropy generation.

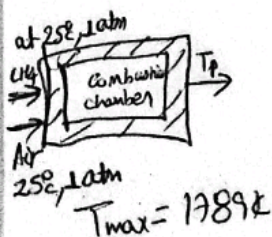
Given

Required

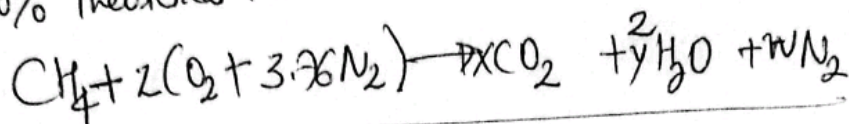
Assumptions

S_{gen}

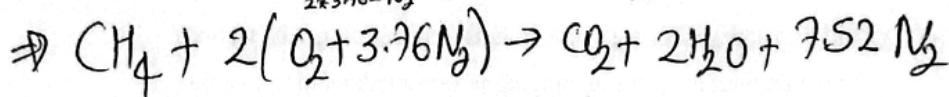
- Steady flow process.
- Adiabatic combustion
- Combustion is complete.
- KE & PE are neglected.
- Air & combustion gases are ideal gases.



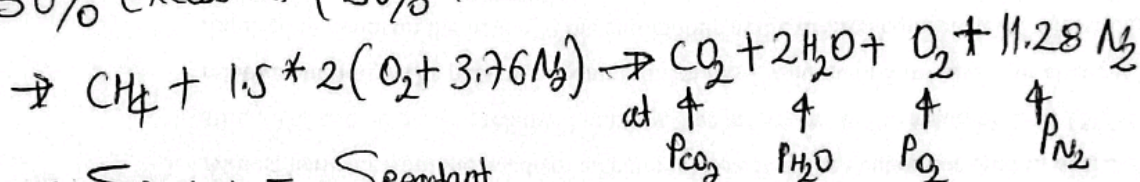
100% Theoretical Air



$$\begin{array}{l|l} C=1 & C=1x \Rightarrow 1x=1 \Rightarrow x=1 \\ H=4 & H_2 \Rightarrow 2y \Rightarrow 2y=4 \\ & y=2 \\ O=4 & O \Rightarrow 2+2=4 \\ & \Rightarrow w \times N_2 = 2 \times 3.76 \times N_2 \\ & w= \end{array}$$



50% Excess air (150% Theoretical air)



$$S_{\text{gen}} = S_{\text{product}} - S_{\text{reactant}}$$

$$\sum_{\text{product}} n_{p,i} \bar{s}_{p,i} - \sum_{\text{reactant}} n_{r,i} \bar{s}_{r,i}$$

$$\Rightarrow S_i = n_i \bar{s}_i(T, p_i)$$

$$= n_i (\bar{s}_i^\circ(T, p_{0@1\text{atm}}) - R_u \ln \frac{y_i p_m}{p_0})$$

$p_0 < 1\text{atm}$

Table A-1 to A-26
like $\bar{h}$

	n_i	n_m	y_i	$\bar{s}_i^\circ(T, 1 \text{ atm})$	$-R_u \ln y_i p_m$	$n_i \bar{s}_i$
				186.16	—	186.16
CH_4	1	1	1		12.975	654.06
O_2	3	14.28	0.21	205.04	1.96	2183.47
N_2	11.28		0.79	191.61		
CO_2	1	15.28	0.0654	302.517	22.674	325.19
H_2O	2		0.1309	258.957	16.905	551.74
O_2	1		0.0654	264.471	22.674	287.15
N_2	11.28		0.7382	247.977	2.524	2825.65
$\sum y_i = 1$						

$S_{\text{Reactant}} = 3023.61$

$S_{\text{Product}} = 3789.71$

$S_{\text{reactant}} = 3023.69$

$S_{\text{product}} = 3789.71$

$$n_m \text{CH}_4 = \frac{1}{2} = y_i$$

$$\text{Air}$$

$$n_m = 1.5 \times 2 \times 3.76 = 11.28$$

$$n_{\text{O}_2} = 3$$

$$n_{\text{N}_2} = 11.28$$

$$n_m \text{ for products} = (1 + 2 + 1 + 11.28) = 15.28$$

$$y_{\text{CO}_2} = \frac{1}{15.28} = 0.0654$$

$$y_{\text{H}_2\text{O}} = \frac{2}{15.28} =$$

$$y_{\text{O}_2} = \frac{1}{15.28} =$$

$$y_{\text{N}_2} = \frac{11.28}{15.28} = 0.738$$

$$\Rightarrow S_{\text{gen}} = S_{\text{product}} - S_{\text{reactant}}$$

$$= 3789.71 - 3023.69$$

$$= 766.02 \text{ kJ/kmol CH}_4$$

$$X_{\text{destroyed}} = T_0 S_{\text{gen}} = (298 \text{ K}) \times 766.02 = 228.28 \text{ MJ/kmol CH}_4$$